

Statistical Physics

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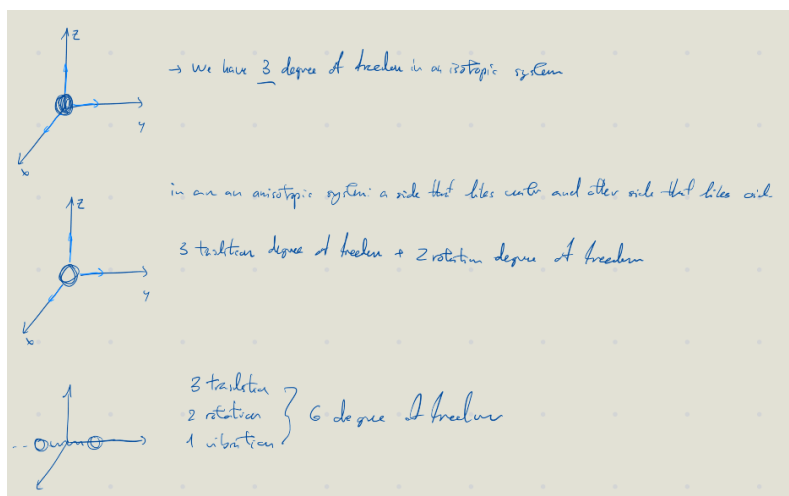
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Capítulo 1

Microscopic description

Statistical physics serves as the bridge that connects the microscopic world of atoms and molecules to the macroscopic phenomena observed in our everyday lives. At its core, this field is concerned with unraveling the mysteries of nature by understanding how countless microscopic particles, when aggregated, give rise to the familiar and predictable behaviour of macroscopic systems. This chapter provides us with the tools and knowledge necessary to comprehend the underlying dynamics of particles at the smallest scales. Here, we shift our focus from macroscopic observables like pressure and temperature to study individual particles – their motions, interactions, and the probabilistic nature of their behaviour. As we walk through the microscopic world, we will uncover the principles governing the behaviour of molecules and atoms. We will decipher the significance of concepts such as phase space, Hamiltonian mechanics, and quantum states, all of which play crucial roles in painting a vivid portrait of the microscopic universe. By the end of this chapter, you will possess a profound understanding of how microscopic details pave the way for macroscopic phenomena.

1.1. Classical Description



1. An isotropic atom has no rotational degrees of freedom, but only three translational degrees of freedom. Since any rotation of a perfectly symmetrical sphere constitutes an identity transformation, the orientation does not alter the particle's interaction potential. Consequently, the system's energy remains invariant under rotation, meaning no thermal energy can be coupled to this motion, and the rotational degrees of freedom are effectively non-existent in the statistical count.
2. An anisotropic atom of two sides, one like water and one like oil has 2 rotational degrees of freedom and the three translational degrees of freedom. In this scenario, where the northern and southern hemispheres exhibit distinct interaction potentials, the perfect spherical symmetry is broken and replaced by axial symmetry. A rotation around the Z-axis (the axis of symmetry) leaves the configuration invariant, meaning it does not change the physical state. However, rotations around the X and Y axes alter the orientation of the hemispheres relative to their surroundings, creating distinguishable states. Consequently, we account for exactly two rotational degrees of freedom.
3. A diatomic molecule has 3 translational, 2 rotational and 1 vibrational (due to the distance between both particles) degrees of freedom.

Just for curiosity

Degrees of Freedom (s)	A (Atom $\Rightarrow$ N=1)	Linear Molecule	Non-linear Molecule
Translational	3(N)	3	3
Rotation	0	2	3
Vibration	0	3N-5	3N-6
Total	3(N)	3N	3N

The degree of freedom is always 3N.

In this chapter we will only treat particles as **simple spheres**.

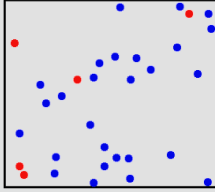
- Let's consider a mechanical system composed of a certain number of particles that has a total of **s degrees of freedom**. To specify the **dynamic state** of this system, it is necessary to know the **positions and velocities** of all the particles that comprise it. Specifically, a total of s position coordinates and s corresponding momenta associated with these coordinates.
- Making use of *Analytical Mechanics*, we can employ generalised coordinates and generalised momenta:
$$\{q_1, q_2, \dots, q_s, p_1, p_2, \dots, p_s\} \equiv \{\mathbf{q}, \mathbf{p}\}$$

Generalised coordinates and momenta can correspond to the **positions** and **momenta** of the particles in the system, although they can also represent more abstract quantities such as **angles**, **wave amplitudes**, or the different **internal degrees of freedom** of a particle.

In classical physics, the description of a system often involves a fundamental and powerful concept known as phase space, here indicated as Γ . Phase space provides a comprehensive mathematical framework for understanding and predicting the behaviour of physical systems. It is particularly essential in classical mechanics for analysing systems with complex configurations and interactions. Phase space provides physicists and engineers with a powerful tool for modelling and analysing

complex systems. It allows us to visualise and study the system's evolution over time by tracing its path through the multidimensional space. This approach simplifies the prediction of a system's future behaviour, making it a fundamental concept in classical mechanics and other branches of physics. Understanding phase space is essential for tackling a wide range of problems, from the motion of celestial bodies to the behaviour of particles in a gas. It forms the basis for classical mechanics and serves as a foundation for more advanced topics in physics. At its core, phase space is a multidimensional mathematical space that offers a complete representation of the state of a physical system. Each point within phase space corresponds to a **unique state of the system**. The number of dimensions in phase space equals the number of **generalised coordinates** and **generalised momenta** required to define the system's configuration. For example, in a system with s degrees of freedom, the corresponding phase space would be a $2s$ -dimensional space. This concept allows us to track the system's evolution through all possible states by examining its trajectory in phase space.

Monoatomic gas composed of N molecules confined in a box.
Degrees of freedom: $s = 3N$



$$\begin{cases} q_1 = x_1 & q_2 = y_1 & q_3 = z_1 \\ q_4 = x_2 & q_5 = y_2 & q_6 = z_2 \\ \dots & \dots & \dots \\ q_{3i-2} = x_i & q_{3i-1} = y_i & q_{3i} = z_i \end{cases}$$

$$\begin{cases} p_1 = m_1 \dot{x}_1 & p_2 = m_1 \dot{y}_1 & p_3 = m_1 \dot{z}_1 \\ p_4 = m_2 \dot{x}_2 & p_5 = m_2 \dot{y}_2 & p_6 = m_2 \dot{z}_2 \\ \dots & \dots & \dots \\ p_{3i-2} = m_i \dot{x}_i & p_{3i-1} = m_i \dot{y}_i & p_{3i} = m_i \dot{z}_i \end{cases}$$

A microstate is defined as the set of coordinates and momenta that completely characterise the dynamic state of a system.

$microstate \equiv \{q, p\}$

Monatomic: only translational degrees of freedom. Diatomic: translational, rotational, and vibrational degrees of freedom.

Consider a monoatomic gas composed of N molecules confined within a box. In this case, the degrees of freedom are given by $s = 3N$. Its generalized coordinates for positions and momenta can be written as follows:

Generalised coordinates

$$\begin{cases} q_1 = x_1 & q_2 = y_1 & q_3 = z_1 \\ q_4 = x_2 & q_5 = y_2 & q_6 = z_2 \\ \dots & \dots & \dots \\ q_{3i-2} = x_i & q_{3i-1} = y_i & q_{3i} = z_i \end{cases}$$

Generalised momenta

$$\begin{cases} p_1 = m_1 \dot{x}_1 & p_2 = m_1 \dot{y}_1 & p_3 = m_1 \dot{z}_1 \\ p_4 = m_2 \dot{x}_2 & p_5 = m_2 \dot{y}_2 & p_6 = m_2 \dot{z}_2 \\ \dots & & \\ p_{3i-2} = m_i \dot{x}_i & p_{3i-1} = m_i \dot{y}_i & p_{3i} = m_i \dot{z}_i \end{cases}$$

In this representation, q_i denotes the position coordinates, while p_i corresponds to the momenta of the i -th particle within the system. These variables, combined with time, allow us to describe the complete state of the system dynamically.

Microstate Definition

A microstate is defined as the set of coordinates and momenta that completely characterise the dynamic state of a system.

$$microstate \equiv \{\mathbf{q}, \mathbf{p}\} \equiv \{q_1, q_2, \dots, p_1, p_2, \dots, p_s\} \quad (1.1)$$

1.1.1. Phase Space

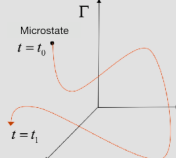
- The phase space, denoted as Γ , is defined as a $2s$ -dimensional metric space, where s is the number of degrees of freedom in the system.
- The axes of this phase space correspond to the variables $\{q_1, \dots, q_s, p_1, \dots, p_s\}$. In the phase space, the **microstate of the system is represented by a point**.
- The temporal evolution of the system results in a **trajectory** within the phase space, traversing different microstates.

Example: diatomic molecule (e.g., O_2)

$s = 3N = 3 \times 2$ degrees of freedom $\Rightarrow \Gamma$ is a space of $2s = 12$ dimensions

Dimensions:

- 3 positions of the center of mass q_x, q_y, q_z
- 3 linear momenta p_x, p_y, p_z
- 2 angular coordinates: θ and ϕ
- 2 angular momenta, p_θ and p_ϕ
- 1 stretching coordinate (distance between atoms), q_r
- 1 stretching momentum: p_r



The space of phases (or phase space), denoted as Γ , is defined as a $2s$ -dimensional metric space, where s is the number of degrees of freedom in the system. The axes of this phase space correspond to the variables $\{q_1, \dots, q_s, p_1, \dots, p_s\}$. In the phase space, the **microstate of the system is represented by a point**. Furthermore, the temporal evolution of the system results in a trajectory within the phase space, traversing different microstates.